

# Jing-Zhang Open Pulse: A Verifiable AI Innovation Public Belt

Package version note: This package is iteration v3.9; v3.3 names the retained spatial-object and drawing-export layer. v3.9 brings the human-first buffer, minimum city-API fields, compute–energy gates and versioned governance to the three nodes with bilingual figures and structured contracts; it changes no spatial objects, metrics, source classifications or implementation boundary.

Core proposition: Jing-Zhang Open Pulse does not string AI devices into a showcase axis. It turns the heritage-park corridor into a civic loop: the three key areas respectively host trusted testing, open conversion and city experience; every component carries an ordinary equivalent service, a human owner, positive and negative evidence, a public receipt and a retirement decision. The proposal carries spatial judgment; GeoJSON, metrics, matrices and 40 gates carry recomputation and audit evidence.

## One-page executive brief: civic loop and one-component/one-proof-mile

- Review question · Readable answer · Primary verification entry ·
- --- · --- · --- ·
- Brief alignment · The proposal serves the taskbook goals of a “global AI industrial highland” and an “AI pilgrimage destination”; three scopes, three key areas, three zones/two wings, three positions, five functions and agent.1–agent.6 then form one belt–three nodes–scenarios–operations chain. Regional interfaces remain conceptual suggestions, not partnership commitments. [source:AGENT-TASKBOOK] · compliance\_matrix.json, taskbook-crosswalk.json, regional-ecosystem.json, scope chapters ·
- Originality · “Civic pulse + One Component, One Proof-Mile”: every scenario has an ordinary equivalent, an accountable owner, positive/negative evidence, a receipt and a retirement decision. · civic-pulse-protocol.json, proof-mile-delivery.json ·
- AI and planning innovation · AI organizes questions, supplies explainable assistance and returns evidence; it does not replace planning, procurement, medical or legal decisions. Spatial moves land in three stations, slow routes, public space and bounded tests. · scenario-operation-matrix.json, key-area-node-plans.json ·
- Implementation feasibility · Move from ordinary service base to bounded prototype, supervised public window, then scale, repair or retire; eight action packages specify owners, dependencies, acceptance and stop rules, while an operational assurance contract makes liability, insurance assumptions, cyber incidents, flood equity and rollback visible in one release record. · operations-matrix.json, construction-readiness.json, operational-assurance-contract.json, phasing chapter ·
- Public interest · Residents, older/disabled people, caregivers, night workers, visitors, merchants, developers and maintainers retain ordinary routes; paper, human and account-free services cannot be replaced by AI. · persona-and-inclusion-matrix.json, public-interest-audit.json ·
- Risk and compliance · Provisional boundaries, unknown baselines, design targets and paper methods are not upgraded into official red lines, existing facts or approvals. Missing rights, fire, heritage, privacy, maintenance, insurance, cyber isolation, flood-equity or takeover proof stops the concept. · sources.json, risk.json, operational-assurance-contract.json, v2-evidence-gate-index.json ·
- Expression completeness · Chinese and English proposals, five fixed review figures, A3/A0, offline HTML, recomputable metrics and three matrices share one spatial/version ledger; v3.3 subdivides buildings, green space, public rooms, scenario anchors, phases and pending constraints without changing the prior area metrics. · manifest.json, metrics.json, geometry/\*.geojson, qa-readiness.json ·
- Source-register completeness · The 68 unique source IDs in the Chinese proposal match the 68 records in package sources.json one-to-one; five are formal-usable, four are background\_only, one is provisional-only, four are package build-toolchain records, one is a boundary cross-check background record, and the rest are repository navigation, case or standard records. Every record provides a title, publisher or registering body, URL/path, access date, authority level, usable scope and disabled scope. data/source\_registry.json remains authoritative for formal/provisional use boundaries; package registration does not upgrade provisional or background material into formal evidence. · sources.json, data/source\_registry.json, visual/assets/evidence-ledger.json, visual/assets/territorial-data-baseline.json ·

## v3.3 Three key areas: spatial moves before technology labels

- Area · Spatial problem · Design move and perceptible deliverable · Release boundary ·
- --- · --- · --- · --- ·
- Zhongzhi Garden AI Accelerator · Qinghe edge, low-carbon exchange, access and embodied-AI tests must not compete for the same public route. · Trusted test garden: connect green space, standards workshop, industry display, low-speed enclosed test court and emergency-stop nodes into a low-disturbance loop back to the park. · Drainage, winter slip, fire, rights, data security and human takeover first; provisional polygons do not substitute for parcel conditions. ·
- Beijing AI Origin Community · Near-campus transfer, talent life, open-source release and quiet daily use must coexist. · Open conversion street: stitch campus, park and neighbourhood movement with a release hall, IP advice, shade service bay and paper receipt point. · Accessibility, night quiet, daily services and human windows come before small events or AI assistance. ·
- Dazhongsi AI Industry Cluster · Station quadrants, enterprise exchange, commercial events and residents returning home can conflict. · Urban experience living room: station walking, four-quadrant crossings, quiet chain, short exhibitions and a capacity-reducible international route form a recoverable public interface. · Fire, crowd, noise, drainage and rail/road coordination remain prerequisites; no event capacity or delivery schedule is promised. ·

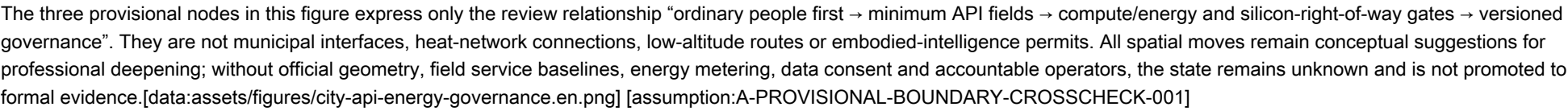
The three areas correspond to PROV-KEY-001, PROV-KEY-002 and PROV-KEY-003; these are provisional design constraints, not official red lines. The moves are conceptual suggestions for professional deepening after formal geometry, rights, controls, fire, heritage and municipal conditions are available.

## v3.9 Human-first buffer and city API: connect the hard gates to three nodes

This iteration compresses “people first, machine-callable, reversible on failure” into a reviewable spatial contract rather than another AI display layer. human-first-buffer.json returns indigenous/older residents, workers at risk of displacement, digitally excluded users and night workers to the three nodes: human windows, paper/telephone service, a skills-retraining and appeal corridor, and screen-free night rest must precede model service. city-api-energy-contract.json then defines asset state, consent, accountable role, human fallback, stop condition and release note as the minimum city-API fields.[data:visual/assets/human-first-buffer.json] [data:visual/assets/city-api-energy-contract.json]

- Human/machine interface · Conceptual spatial move · Review gate and data gap ·
- --- · --- · --- ·
- Human-first buffer · At AI Origin retain a staffed registration desk, intergenerational learning and a skills-retraining corridor; at Dazhongsi reconnect night human service, quiet rest and screen-free green space to the ordinary slow route · Record, but do not claim, resident\_continuity\_retention\_rate, skills\_retraining\_corridor\_completion\_rate and manual\_channel\_availability\_rate; resident, merchant, illuminance, noise, accessibility and night-service baselines are missing [data:visual/assets/human-first-buffer.json] [metric:public\_space\_ratio] ·
- City API layer · At Zhongzhi provide an asset/geometry-version observation desk; across the three nodes show consent, responsible role, human fallback and stop condition. Machines call only auditable minimum fields; people and ordinary services do not depend on an account · Every window requires asset\_id, geometry\_version, uncertainty, human\_fallback and stop\_condition; consent, interface responsibility, network isolation and public receipt remain for professional verification [data:visual/assets/city-api-energy-contract.json] [depth:versioned\_governance] ·
- Compute–energy and versioned governance · PUE tiers, renewable/heat interfaces, human–machine movement and the Xiaoyue River Wing embodied-test candidate are approval-dependent concept gates; annual review, public receipt, release note and rollback connect an AI three-month iteration to long-term urban maintenance · PUE, renewable power, heat-network, low-altitude or test permits remain project-level unknown; they are not settled policy, engineering feasibility or capacity commitments. Any failed fallback, unanswered resident appeal or missing responsibility freezes scale and restores ordinary service [data:visual/assets/city-api-energy-contract.json] [standard:BEIJING-DATACENTER-ENERGY-REVIEW-2023] ·

Conceptual suggestion / provisional constraint / unknown blocks release; for professional refinement



1. Review: lock the official/provisional geometry version, field baseline, rights and accountable owner; if a critical input is missing, keep the state unknown.
2. Base: provide physical wayfinding, paper maps, human windows, seating/shade, water, drainage and account-free help first; ordinary service never waits for model recovery.
3. Test: operate only in supervised, time-bounded, low-speed windows with emergency stop and human takeover; record positive, negative, near-miss, complaint and group-difference evidence.
4. Decide: independent reviewers compare retain, repair, scale or retire; no scenario result is inherited by another component.
5. Receipt: publish version, sources, limits, deletion/withdrawal and maintenance records; on failure freeze scale and restore passive service.

First auditable slice: S02 synthetic low-speed delivery-robot rehearsal (contract only; not run): anchor it to the provisional geometry/roads.geojson#ROAD-001 segment, with one synthetic device run, one human spotter and one marked return bay; no live delivery and no public-road release. The accessible route stays open, and any pedestrian conflict, route obstruction, stop-signal failure, loss of spotter sightline or unavailable return bay stops the rehearsal immediately. Paper notice, visible stop state, non-digital explanation and a hand-cart/human-delivery equivalent are release prerequisites. Record only a synthetic run ID, route window, yield/stop/bay-return events; collect no personal data. The current record remains result\_status=not\_run and release\_decision=hold. Full fields are in visual/assets/example-s02-embodied-test-window.json; this compresses a reviewable spatial window, human decision rights, data minimisation and restoration into one auditable record, not robot performance, public acceptance or deployment approval.

The full formal dossier follows with design basis, three scopes, key areas, ecosystem and scenarios, land use, transport, municipal and blue-green systems, implementation, metrics, risk and references. Complete sources, standards, depth, data and stop rules remain in structured attachments.

This proposal follows the official open-call announcement, the repository site package and the agent taskbook. The machine-readable evidence chain is registered in `sources.json`, `standard_matrix.json`, `design_depth_matrix.json`, `compliance_matrix.json`, and `visual/assets/evidence-ledger.json`. The most important source IDs are `[source:OFFICIAL-ANNOUNCEMENT]`, `[source:AGENT-TASKBOOK]`, `[source:SITE-PACKAGE]`, `[source:SOURCE-REGISTRY]`, `[source:PROCESSED-FACT-PACK]`, `[source:BOUNDARY-SOURCE]`, and `[source:KEY-AREA-SOURCE]`.

## Provisional boundary cross-check: preserve the discrepancy as a risk, not a red line

Administrative-scale evidence: calibrate questions, do not allocate space

To keep a strong audit package connected to real conditions, this version adds four official public background sources. All remain background\_only: they translate administrative-scale observations into service interfaces and corridor-level data-collection questions, but do not enter spatial-control metrics, replace field baselines, or become project

targets.[source:HAIDIAN-2025-STATISTICAL-BULLETIN] [source:NBS-2025-STATISTICAL-COMMUNIQUE]

The transport and university-service records are kept as separate interface calibrations; city-level travel shares and one event's attendance are not converted into spatial capacity, footfall or participation commitments.[source:BJTRC-2026-TRANSPORT-REPORT] [source:HAIDIAN-37-UNIVERSITIES-SERVICE-NEEDS] [data:visual/assets/territorial-data-baseline.json#baseline-01-haidian-innovation-and-service]

· Evidence scale and public observation · Spatial/operational translation · What it cannot prove ·  
· --- · --- · --- ·

· In 2025 Haidian reported 92 national key laboratories, 123 registered large models, 599 high-value invention patents per 10,000 people, 57,900 technology contracts and CNY 405.31 billion in contract value; it also reported 1,456 health institutions and 239 community health centres/stations.[source:HAIDIAN-2025-STATISTICAL-BULLETIN] · The three nodes are not generic incubator displays: Zhongzhi prioritises standards/model assurance, AI Origin prioritises conversion, rights/IP and public-service clinics, and Dazhongsi prioritises reversible urban experience and public receipts. · These totals do not locate resources in the corridor, prove participation, define footfall or establish facility capacity; health statistics do not support diagnosis or facility siting. ·

· National 2025 statistics report 16 high-value invention patents per 10,000 people, CNY 7,573.4 billion in technology-contract value and CNY 3,926.2 billion in R&D expenditure.[source:NBS-2025-STATISTICAL-COMMUNIQUE] · Use this only as a scale check for the innovation narrative; judge the proposal by assurance quality, accountability, negative evidence and reuse rather than ranking or growth promises. · Administrative levels, institutional coverage and statistical definitions are not assumed to be directly comparable, and no Haidian/Jing-Zhang performance target is derived. ·

· Beijing central-city 2025 statistics report about 39.24 million weekday trips, 76.5% green travel, and rail/bus/bicycle shares of 15.0%/8.9%/22.3%. [source:BJTRC-2026-TRANSPORT-REPORT] · Put rail transfer, walking and cycling first at the public interface; test the last 300–800 metres with a small, time-bounded, staffed and reversible window before any event or route expansion. · It does not provide Jing-Zhang station OD, crossing safety, parking demand, capacity or road-engineering conclusions; OSM and city statistics must be replaced by corridor surveys. ·

· One public Haidian university-service event involved 37 resident universities and approximately 130 participants.[source:HAIDIAN-37-UNIVERSITIES-SERVICE-NEEDS] · Give AI Origin a proposed appointment-based conversion clinic for laboratory/equipment safety, rights/IP, asset operation and results transfer, with human and paper alternatives. · One event does not prove that all 37 universities share the same demand, have agreed to participate, or define a service volume. ·

Source URLs, access dates, statistical scope, observations, transformations and prohibited inferences are recorded in sources.json and visual/assets/territorial-data-baseline.json. The chain is therefore background fact → design move → missing corridor evidence, not administrative statistics disguised as spatial conditions.[data:visual/assets/territorial-data-baseline.json#baseline-03-transport-context]

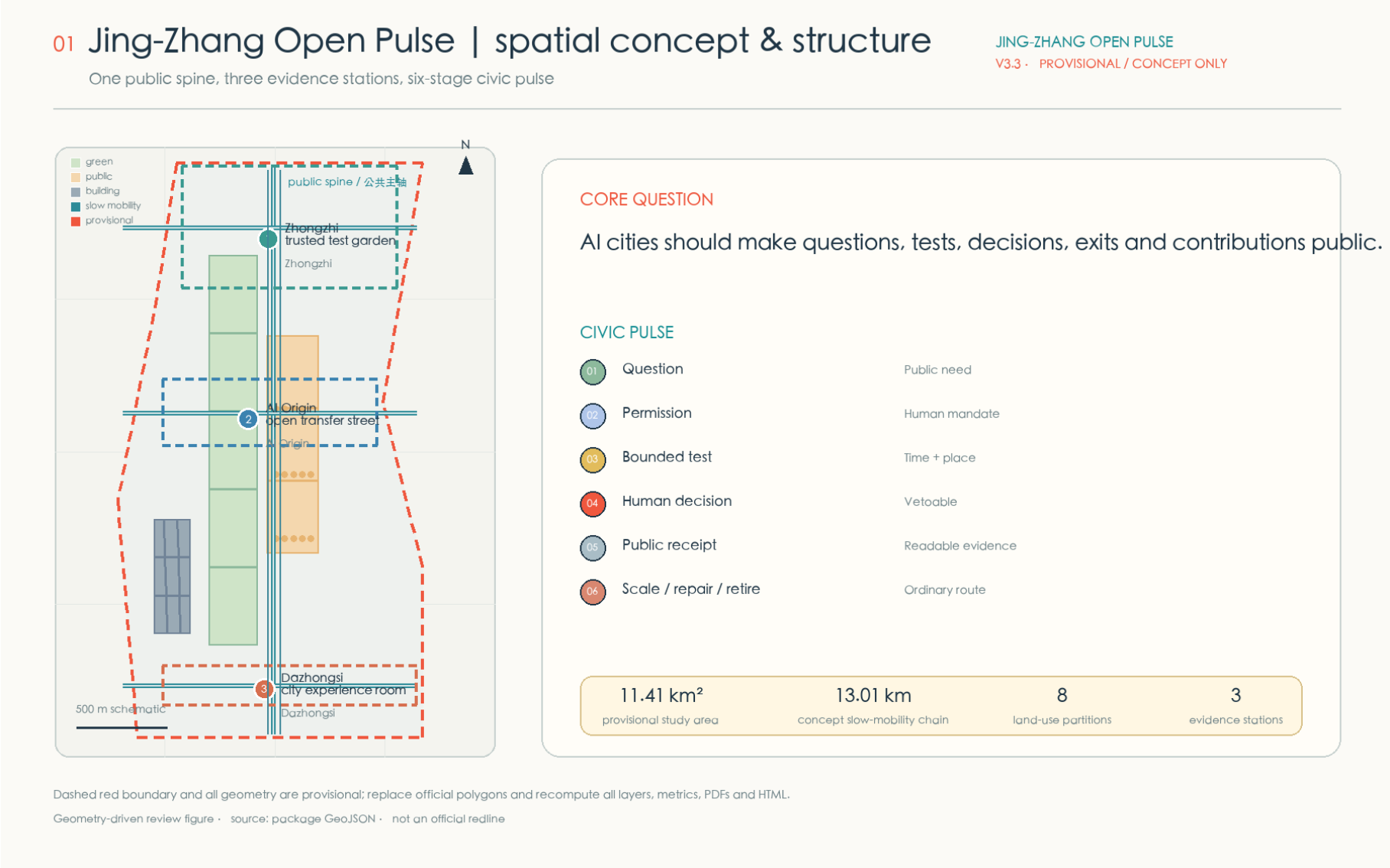
Three-Level Scope Framework

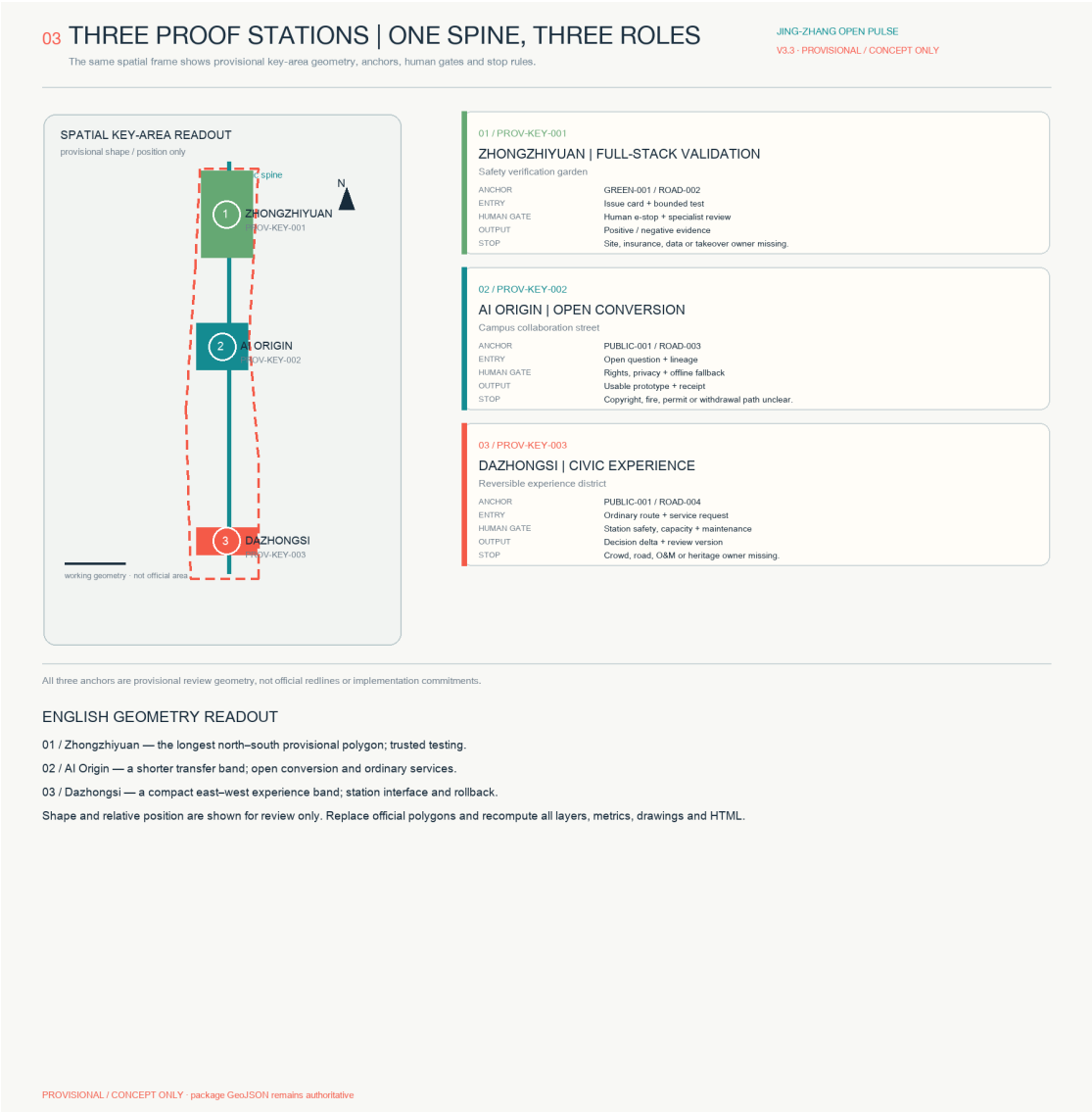
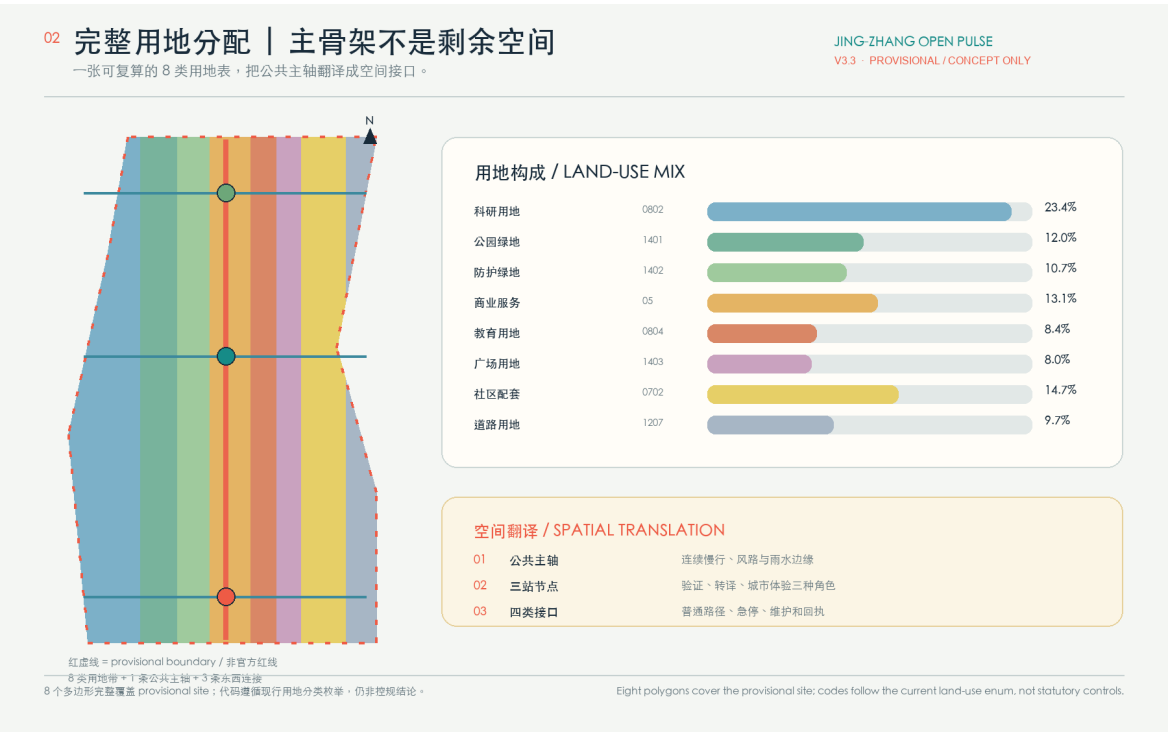
The proposal works at three linked scales: the 43.6 km² coordinated research area, the 11.4 km² overall design area around the Jing-Zhang heritage park, and the 368.4 ha detailed-design area containing three key areas. These figures are taskbook scope descriptors, not a claim that the current provisional polygons reproduce official measurements. The three levels are cross-referenced in compliance\_matrix.json and visual/assets/taskbook-crosswalk.json.

The named spatial concept is Jing-Zhang Open Pulse: one belt, three nodes, many scenes, and a blue-green walking loop. “One belt” is a design method rather than a new legal boundary; the three nodes correspond to the three taskbook key areas; the loop joins slow movement, public space, water, shade and everyday services.

· Scale · Design question · Evidence and decision ·  
· --- · --- · --- ·

- Coordinated research · How can knowledge, talent, compute, capital and public demand circulate? · Eight-mechanism regional loop; conceptual only. ·
- Overall design · How do renewal, industry, transport, municipal systems and character fit together? · Linked geometry layers, metrics and phasing. ·
- Key areas · How does each area become experienced, operated and reversible? · Node concepts, 14 scenario rows and component library. ·





## Coordinated Research Area: Industry and Future City Research

The research-area strategy is a public-facing innovation chain: university and community demand → open collaboration → enterprise conversion → ordinary public experience → evidence and international exchange. It is not an assertion that any named institution has agreed to participate. The regional mechanism diagram makes the proposed interfaces inspectable: Haidian universities and communities originate questions; Huairou contributes upstream science; Future Science City contributes engineering and energy testing; Beijing E-Town contributes manufacturing and deployment; the Jing-Zhang belt provides bounded public test windows; the broader Beijing-Tianjin-Hebei network compares and diffuses results.

The eight mechanisms are land, space, industry, capital, talent, compute, data and scenario. Each mechanism has an inbound need, an outbound return and an accountable future operator in visual/assets/regional-ecosystem.json. No partnership, funding, land allocation, or government commitment is asserted. The regional return loop is: originate → engineer → book → test → publish evidence → scale or retire.

The case comparison now uses six officially sourced public examples as mechanism references, not as local performance claims. The official pages are listed in the table below.

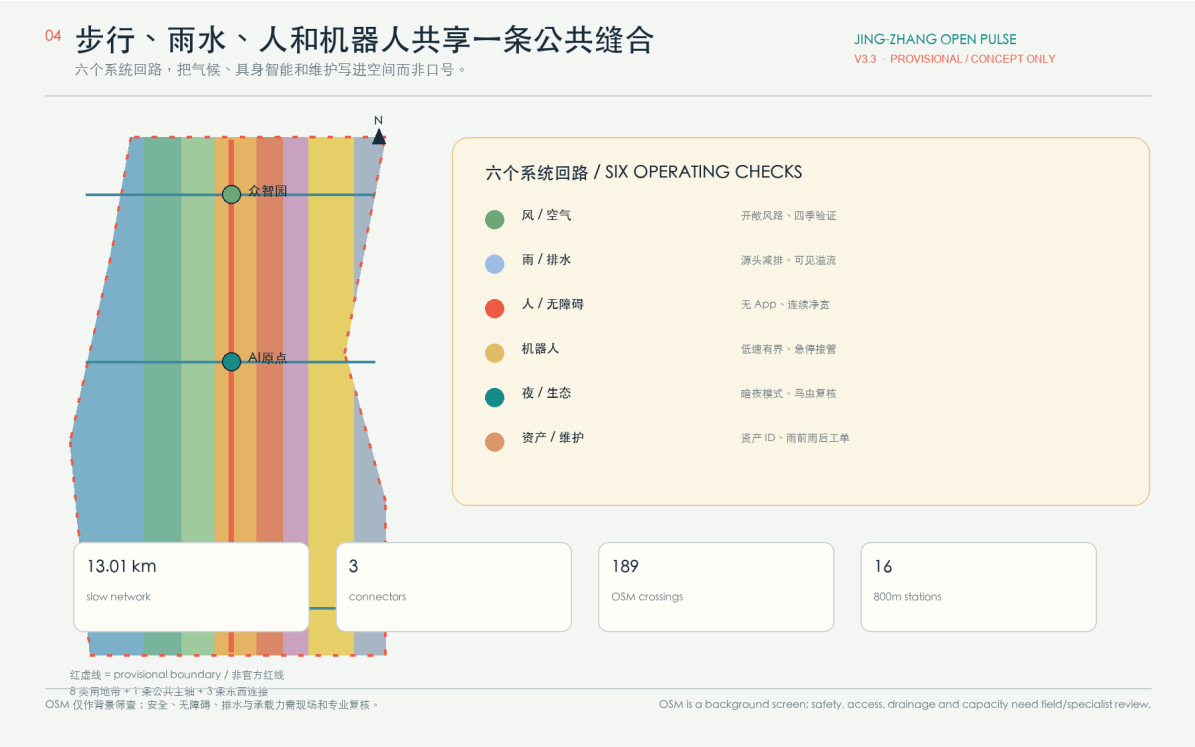
· Official example · Verified mechanism · Jing-Zhang interface · Boundary · Source ·

· --- · --- · --- · --- · --- ·

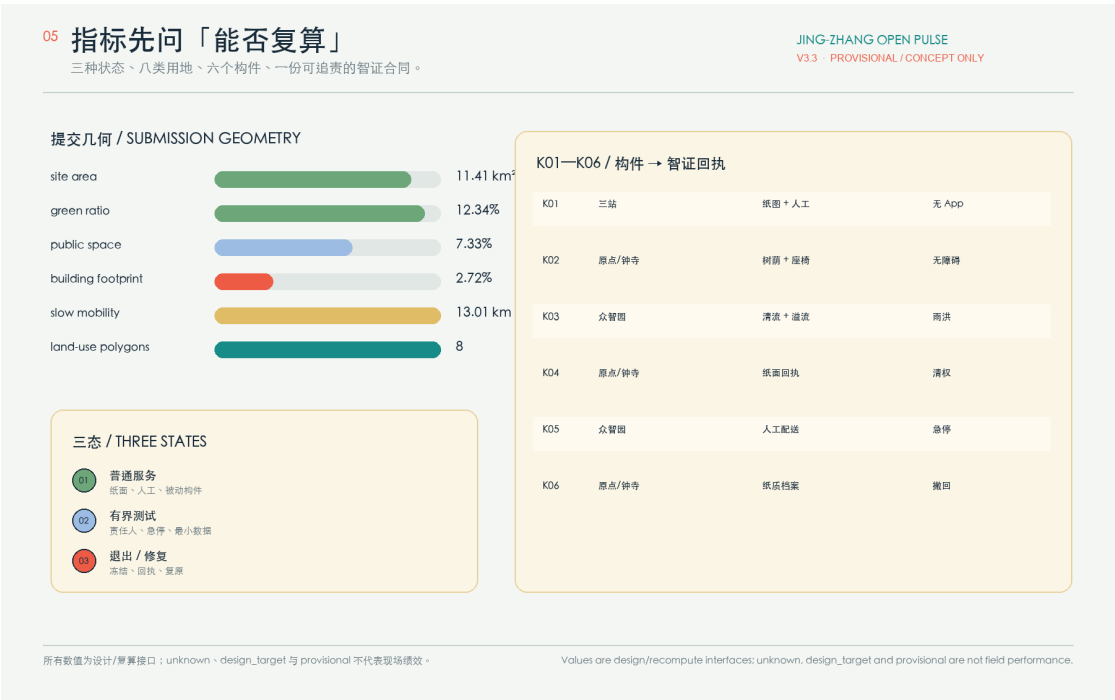
- Helsinki AI Register · Public inventory and feedback route for city AI systems · Status board with purpose, data boundary, accountable lead, state and complaint route · Does not replace safety, accessibility, procurement or consent review · [source:CASE-HELSINKI-AI-REGISTER] ·
- Amsterdam Algorithm Register · Plain-language public explanation of municipal algorithm use · Preflight record linked to review, stop and redress paths · Not Chinese legal compliance; protected details require review · [source:CASE-AMSTERDAM-ALGORITHM-REGISTER] ·
- Singapore AI Verify · Standardised AI testing, open-source collaboration and assurance sandbox · Booked human-led assurance window with model card, test note and rollback · Not certification, procurement approval or permission to test residents · [source:CASE-SINGAPORE-AI-VERIFY] ·
- Decidim Barcelona · Traceable digital and face-to-face participation infrastructure · Contribution wall plus paper, walk-in and multilingual routes; publish what changed · Digital participation cannot replace access for people without accounts or devices · [source:CASE-DECIDIM-BARCELONA] ·
- UK ATRS · Standardised public record of why and how algorithmic tools are used · Compact record for purpose, owner, data, human review, alternatives, limits and incidents · Not a Beijing statutory requirement; local legal review remains necessary · [source:CASE-UK-ATRS] ·
- Seoul AI Foundation · Institutional capacity linking research, public-service application, talent and global exchange · Named interface among research, talent, public service and exchange · No partnership, funding or city mandate is implied · [source:CASE-SEOUL-AI-FOUNDATION] ·

The 68 source IDs, access dates and use boundaries are recorded in sources.json. This is a mechanism comparison, not proof that Jing-Zhang has adopted or partnered with any of these organisations. The original six design patterns remain in visual/assets/case-mechanism-matrix.json as the local test layer.

The policy and enterprise-development interface is recorded separately in visual/assets/case-policy-enterprise-crosswalk.json: every case is linked to a policy tool, an enterprise question, a local scenario, acceptance evidence and a do-not-copy boundary. This moves the comparison from “seen a case” to “how a company enters a bounded testing, procurement, service and review loop” without turning a design suggestion into enterprise recruitment, funding or government commitment. visual/assets/policy-enterprise-playbook.json turns that interface into 42 growth-stage cards, beginning with a public AI register and feedback card; visual/assets/industry-validation-cases.json adds four bounded validation windows; and visual/assets/landmark-honor-crosswalk.json links four public responsibility/contribution markers to the policy and enterprise interfaces. These are conceptual protocols, not procurement, funding or partnership commitments.



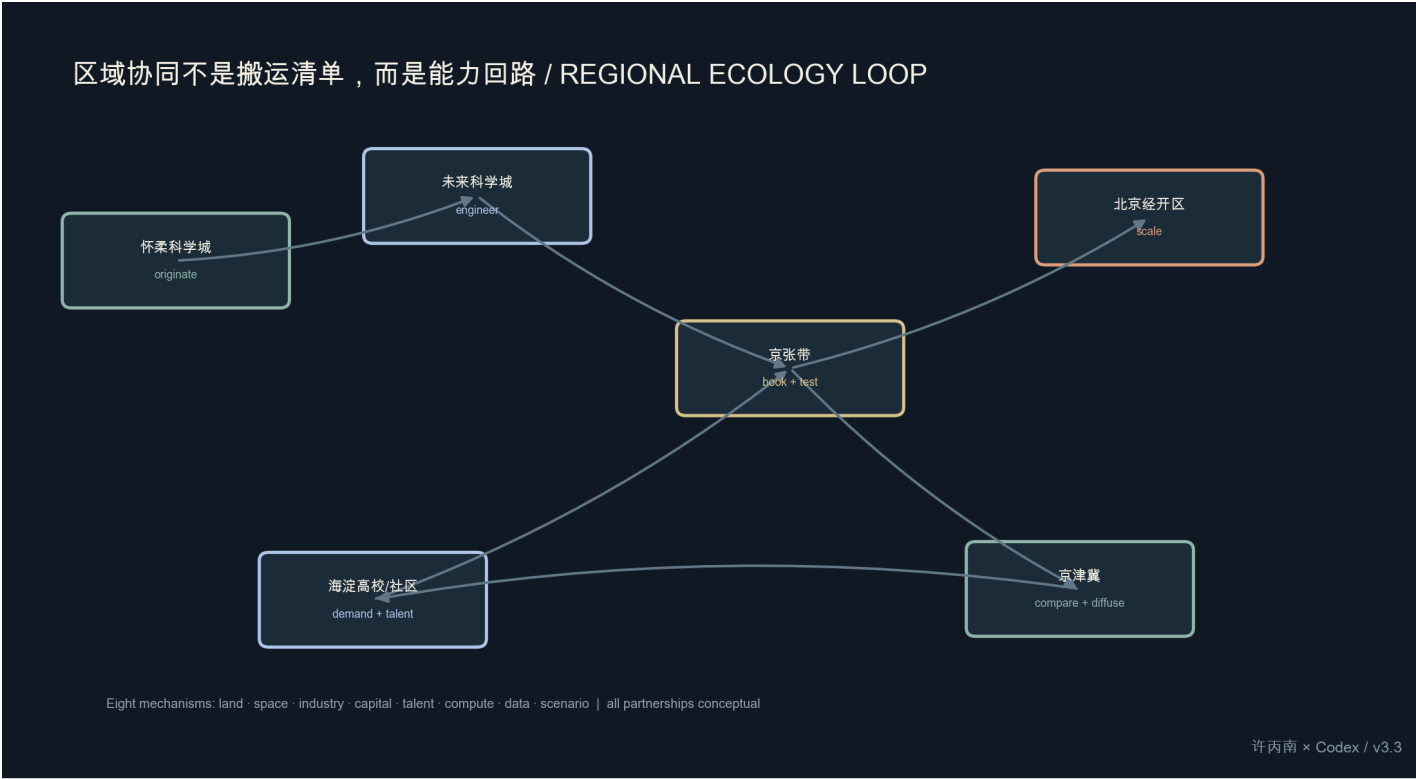




English translation of figure text

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05 / Metrics first: can it be recomputed?  
Three states, eight land uses, six components, one auditable evidence contract.  
Submission geometry: site area 11.41 km²; green ratio 12.34%; public space 7.33%; building footprint 2.72%; slow mobility 13.01 km; land-use polygons 8.  
K01–K06 / component → evidence receipt: K01 three stations → paper map + human → no app; K02 AI Origin / Dazhongsi → tree pit + seat → step-free; K03 Zhongzhiyuan → clear flow + backflow → rain; K04 AI Origin / Dazhongsi → paper receipt → Qinghe; K05 Zhongzhiyuan → human dispatch → e-stop; K06 AI Origin / Dazhongsi → paper archive → withdrawal.  
Three states: ordinary service = paper, human, passive; bounded test = responsible person, e-stop, minimal data; exit / restore = freeze, receipt, restoration.  
All values are design and recompute interfaces; unknown, design\_target and provisional do not represent field performance.



English translation of figure text

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Regional coordination is a capability loop, not a transfer list.  
Huairou Science City → originate. Future Science City → engineer. Jing-Zhang belt → book + test. Beijing E-Town → scale. Haidian university / community → demand + talent. Beijing–Tianjin–Hebei → compare + diffuse.  
Eight mechanisms: land · space · industry · capital · talent · compute · data · scenario. All partnerships are conceptual and require separate ownership, funding, safety, rights and implementation review.

Overall Design Area: Urban Renewal and Regulatory-Plan-Level Urban Design

The overall design uses a renewal-first structure. Low-efficiency spaces are not assigned invented floor-area ratios or heights; instead, they receive a retain–renovate–demolish decision gate and a requirement for later official-control confirmation. In v3.3, the land-use layer remains an eight-class, non-overlapping design partition with [metric:land\_use\_polygon\_count] polygons and recomputable shares, while the geometry package subdivides the unchanged building union into [metric:building\_unit\_count] conceptual units, the green corridor into [metric:green\_segment\_count] connected segments, and the public interface into [metric:public\_space\_polygon\_count] rooms plus [metric:public\_scenario\_node\_count] scenario anchors. metrics.json exposes these counts and the unchanged area ratios; all polygons and points remain provisional and non-statutory. Roads, constraints and [data:geometry/phasing.geojson#PHASE-001] express the mobility network, pending professional inputs and conditional phasing.

Transport and municipal systems are designed as a continuous human network: rail-station interfaces, step-free crossings, low-speed logistics windows, cycle parking, emergency access, rainwater tree pits, shade, drainage maintenance and ordinary public use. AI may assist observation or scheduling, but a resident must still be able to walk, ask, complain, receive service and leave without an app, account or biometric identification.

Detailed Design of Key Areas

1. Zhongzhi Garden: autonomous-innovation accelerator

Zhongzhi Garden is a garden-like full-stack innovation district. Its public edge displays safety cases, standards work and low-carbon computing rather than turning the park into a private campus. A “stitch threshold” creates a step-free, audible and tactile crossing; a status board shows the current test window, accountable operator, stop condition and non-AI equivalent.

2. Beijing AI Origin Community

The Origin Community is a near-campus conversion and talent neighbourhood. The design stitches campus, park, enterprise and residential routes; creates an open publishing room; and reserves ordinary living, rest and care spaces. Results are presented as cleared models, diagrams and explainers, not as unlicensed historical or corporate imagery.

3. Dazhongsi AI industry cluster

Dazhongsi is an urban, internationally legible AI service and enterprise district. The station interface, four-quadrant pedestrian connections and retail/service edges are treated as one public sequence. Intelligent-device demonstrations are bounded by capacity, fire safety, resident access and a thirty-minute return-to-normal rule after an event.

Each key-area polygon is provisional and each node proposal is a concept for professional refinement. The node-level decision record is mapped to [data:geometry/key\_areas.geojson#PROV-KEY-001], #PROV-KEY-002, and #PROV-KEY-003.

Public Interfaces and Reversible Relationships for the Three Key Areas

The three key areas already have functions and spatial actions, but ordinary routes, staffed services, and equipment back-of-house still need a professional comparison entry. The ledger below gives public-interface and reversible-service relationships only; it does not change the geometry or turn provisional areas into regulatory controls.

- Key area · Public ground-floor choice · Reversible service relationship · First professional evidence to collect ·
- --- · --- · --- · --- ·
- Zhongzhiyuan AI Autonomous Innovation Acceleration Zone · Layer Qinghe public space, R&D display, cycle shelter and enclosed testing; do not expose models or personal data at the public edge · Public observer seat and staffed front desk stay adjacent; equipment, maintenance, and testing back-of-house can close without cutting the walking spine · Existing buildings, ownership, flood/blue-line, fire, energy and enterprise OD ·
- Beijing AI Origin Community · Stitch campus, enterprise and neighbourhood walking; put open publishing, talent service and human entry at an accessible ground floor · Paper, phone, and staffed entries face the ordinary route; test components remain removable and pausable, and resident service is not account-dependent · Accessibility walk, resident/student service baseline, ownership, care and night safety ·
- Dazhongsi AI Industry Cluster · Separate station, four-quadrant walking, international presentation and retail/service edges; keep data display away from quiet residential frontage · Rail arrival, quiet route, and service front desk are separated; events and data display can withdraw without occupying daily resident movement · Station flow, road/curb counts, fire, municipal, ownership and event-day organisation ·

These relationships only identify which public and reversible conditions to compare first. They do not answer where construction is permitted, who holds the space, or what final control parameters should be. A formal scheme must be determined by a professional team after regulatory-plan, survey, ownership, fire, structural, municipal, traffic and public-participation evidence is available [depth:height\_massing\_character] [depth:development\_intensity\_controls].

Until then, renewal prioritises wayfinding, staffed service, cycle shelter, weather seats, accessible ramps and movable equipment; no road redline, building increment or demolition conclusion follows [standard:MOHURD-CONTROL-DETAILED-PLANNING].

AI Innovation Ecosystem, Personas, and AI+ Scenarios

The city is designed for more than one “AI user”. Personas include a university researcher, an SME founder, a community service worker, an older resident or carer, a visitor, a night-shift worker, an accessibility reviewer and a maintenance crew. Every persona has a non-digital fallback, a place to rest, a right to opt out and a visible complaint path.

The eight non-identifying public-user lenses are recorded in visual/assets/persona-and-inclusion-matrix.json: older/resident users; wheelchair, cane and guide-dog users; carers and children; night-shift workers; visitors; small businesses and community-service workers; developers/startups/university teams; and maintenance/operators. user\_persona\_count is a design-target count, not a population statistic or eligibility system. Each lens has a spatial response, offline fallback, test and safeguard, and the package prohibits facial recognition, individual trajectory retention, biometrics and commercial profiling. The matrix rows P-01 through P-08 remain conceptual until informed, opt-out and compensable participation is completed.

The package contains 14 scenario cards in visual/assets/scenario-operation-matrix.json. They cover walking-gap diagnosis, low-speed delivery robots, autonomous-mobility assistance, model red-teaming, open publishing, Jing-Zhang cultural wayfinding, health-service navigation, enterprise service, data-governance demonstrations, intelligent retail, public-safety review, an international AI week walk, rainwater/tree-pit maintenance, and a low-light quiet chain.

The matrix is a design-control instrument, not a promise of service levels. Every row states: spatial carrier, trigger, minimum data, RACI, response/recovery SLA, relative cost band, data retention, non-AI equivalent, stop condition, success metric and phase. The universal gates are: no digital precondition; human review cannot be removed; an accessible route remains continuous; a public status and complaint path exists; and stop/rollback has been tested.

To make a scenario card reproducible rather than merely readable, this package adds the independently designed visual/assets/open-pulse-test-window-record.schema.json and visual/assets/example-s02-embodied-test-window.json. The latter is a fully synthetic S02 low-speed-delivery sandbox test-window record: it puts the provisional road reference, no-app human alternative, minimum fields, visible stop and human spotter, public complaint route, maintenance responsibility and exit-asset plan in one record. Its result status remains not\_run; the record is not robot performance or field acceptance. Any real pilot must create a new record and pass accessibility, safety, maintenance and public-review gates.

To keep that record from remaining merely readable, this iteration adds visual/assets/open-pulse-tabletop-contract.json, open-pulse-tabletop-evidence.json and a dependency-free runner. node visual/assets/run-open-pulse-tabletop.js --check reads only the bundled synthetic S02 record and four branch fixtures: an unprofessionally verified provisional route, a pedestrian conflict or narrowing, a missing spotter or emergency stop, and a review close with no field result. It replays six checks and five rollback steps, preserving paper/staffed/hand-cart ordinary service, clearing the accessible route, publishing hold/withdrawal and deleting only temporary fixture state. To keep the runner from checking only aggregate strings, each of the six acceptance checks now binds to fixture\_ids, scenario\_ids or boundary\_fields; each rollback step also has a unique RB ID with fixture and acceptance-check references. The runner requires 4/4 fixtures, 1/1 scenario, 5/5 boundary fields, 5/5 rollback steps and six unique check IDs to resolve. It also sends each fixture state through a deterministic negative replay, yielding decision\_class=reject\_or\_stop for hold, stop, withdrawal, or temporary-state deletion, and adds an ordinary-route control replay to confirm that a non-trigger input remains continue\_ordinary\_route rather than being rejected. The evidence is 4/4, 6/6, 5/5, 5/5, negative replay 4/4, and control replay continue, while result\_status=not\_run, performance\_results=null and operational\_status=not\_authorized\_not\_run remain explicit. PASS proves only that stop, withdrawal, restoration and non-trigger discrimination are inspectable; it does not prove robot performance, accessibility, public acceptance, staffing, safety or permission [data:visual/assets/open-pulse-tabletop-contract.json] [data:visual/assets/open-pulse-tabletop-evidence.json] [data:visual/assets/run-open-pulse-tabletop.js].

To address the structural-provenance and licence boundary exposed by #706, this version removes the peer-derived Relay Receipt files and replaces them with an independently designed Open Pulse bounded test-window record. The new record is organized around window, place\_window, public\_interface, human\_control, data\_contract, observation, release\_decision and restoration; it does not reuse the #426 schema skeleton. It remains a concept-only review interface, not deployment permission or field evidence.



English translation of figure text

The source graphic is preserved above; this panel translates its detected text.

A full lifecycle for each public token / TOKEN → TEST → RETURN

01 Request · public request

02 Review · data + access + RACI

03 Operate · bounded window + human lead

04 Review · SLA + complaint + evidence

05 Return · ordinary public use

RACI = owner + accountable lead; SLA = response + recovery clock; COST = relative band only; DATA = minimum + retention; STOP = rollback + non-AI equivalent.

14 scenario rows in scenario-operation-matrix.json. Thresholds are design gates, not statutory service levels.

Four industry validation windows: evidence before procurement or scale-up

visual/assets/industry-validation-cases.json defines four reversible, staffed validation windows for model safety, enterprise-service data governance, low-speed embodied AI and multi-channel civic participation. Each couples an enterprise question with a policy interface, acceptance evidence, a stop condition and a non-AI equivalent. The first window asks whether a supplier can reproduce safety, provenance, human review and rollback evidence; the second asks whether enterprise service can shorten a public process without unauthorised data or a binding AI decision; the third asks whether a robot can coexist with an accessible public route before scale-up; the fourth asks whether a civic-service team can reconcile digital, paper, walk-in and multilingual feedback without making an account or device a condition of access. Two independent reviewers, a working manual route, clear rights, visible stop/takeover and a return to ordinary use are release gates. Unclear provenance, privacy spill, discriminatory routing, blocked accessibility, a severe near miss or overdue maintenance stops the window. These are conceptual test protocols, not local-company performance, procurement approval or deployment fact. [source:AGENT-TASKBOOK]

Cultural narrative, public markers, and long-term operations

The cultural narrative uses engineering proof, open collaboration and public reciprocity as a three-part experience. Four non-glitzy markers are proposed: the Century Engineering Questions Wall, the Open-Source Contribution Archive, the Urban Agent Responsibility Desk, and the Enterprise Safety-Governance Responsibility Desk. visual/assets/landmark-honor-crosswalk.json binds each to a spatial carrier, acceptance test, policy/enterprise interface and clearance boundary. They are conceptual nodes for heritage, public-art, management and rights review—not institutional endorsements, advertising sites or an approved event programme.

Node-level concepts and public-interest audit

visual/assets/key-area-node-plans.json makes the three key areas reviewable as sequences rather than slogans: arrival → status board → component → bounded operation → return to ordinary public use. Each node has components, an acceptance test and a professional gate for official parcels, ownership, roads, fire, ecology, heritage, drainage and municipal systems. It is a provisional concept for professional refinement, not an official parcel or construction drawing.

visual/assets/public-interest-audit.json deliberately leaves the baseline unknown until a survey is run. Older residents, disabled people, carers, night workers, children and guardians, visitors, small businesses and maintenance crews are named as participants. The audit tests step-free continuity, service without an app, complaint response, group differences and night light/noise. Participation has paper, staffed and multilingual channels; conflict resolution pauses the scenario, shows the disputed evidence, offers a non-AI equivalent and publishes remedy or withdrawal.

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

The land-use layer forms a complete, closed and non-overlapping eight-class provisional design partition based on the published classification standard. The building layer distinguishes retain, renovate, renew, new-build and to-be-confirmed objects, and records the proposed level of control for building footprint, use, character, massing and height in the matrices. Existing buildings, ownership, regulatory plans and engineering conditions are incomplete; therefore the retain-renovate-demolish register records a method and a calibration checklist, not predetermined conclusions.

Land-use classification follows the published classification standard [standard:MNR-LAND-USE-CLASSIFICATION-GUIDE]. Building height, massing, interface and character controls are managed by [depth:height\_massing\_character], while the retain-renovate-demolish method is managed by [depth:retain\_renovate\_demolish].

The principal land-use and building evidence is carried by [data:geometry/land\_use.geojson#LU-001], [data:geometry/buildings.geojson#BLDG-001] and [metric:building\_footprint\_area\_sqm].

Development-control indicators must remain consistent with metrics.json and the geometry layers. Total floor area, density, setbacks and building-control lines are marked unknown or pending\_control because official conditions are missing; no fixed number is used to manufacture precision. The submitted A3 booklet contains the renewal-project list and indicator review table, the A0 boards foreground the key spatial structure and areas, and the offline HTML provides the linked indicator and layer audit entry.

Transport, Rail, Municipal Infrastructure, and Public Services

The central transport judgement is that the corridor has relatively rich rail resources but a higher risk of discontinuity in the final 300–800 metres. A provisional-area and 800-metre-buffer background screen identifies 16 deduplicated station names and 189 mapped crossing nodes; mapped walking-support lines, cycleways and major road-centreline density are approximately 16.53, 1.06 and 5.47 km/km² respectively. These low-confidence results only set field-survey priorities. They do not establish official station entrances, road red lines, continuous cycling quality or capacity. [source:OSM-TRANSPORT-CONTEXT] [metric:osm\_mapped\_station\_names\_within\_800m\_count] [metric:osm\_mapped\_crossing\_count]

This iteration corrects a directional error in the prior spatial evidence: ROAD-001 was previously only an approximately 1.1-kilometre east–west line and could not support a nine-kilometre innovation belt. v1.3 changes it to an approximately 9.60-kilometre north–south public slow-mobility spine, with east–west stitching branches at the three key areas.



[source:JINGZHANG-FUTURE-BELT-2026] [source:BEIJING-SLOW-MOBILITY] [data:geometry/roads.geojson#ROAD-001]

Branch and length checks return to [data:geometry/roads.geojson#ROAD-002], [data:geometry/roads.geojson#ROAD-003] and [data:geometry/roads.geojson#ROAD-004]. The conceptual network indicators are [metric:design\_north\_south\_spine\_length\_m], [metric:design\_east\_west\_connector\_count] and [metric:design\_slow\_mobility\_network\_length\_m].

The spine and branches describe a network relationship, not a new road, red line or engineering alignment. The North Fifth Ring/Qinghe interface, campus–park–community edges, the four quadrants of Dazhongsi Station, and the North Third Ring–Jingbao Road–Zhichun Road nodes require separate transport, ownership, accessibility and engineering studies. Published public-feedback material confirms that the corridor contains complex interchange and land-use conditions; one concept diagram cannot replace professional judgement. [source:JINGZHANG-PUBLIC-FEEDBACK] [depth:traffic\_rail\_slow\_parking]

Resident transport experience is evaluated through six lenses: walking continuity, crossing safety, cycling comfort, rail transfer, night-time comfort and 15-minute daily services. The local baseline is missing, so the satisfaction metric remains unknown; the pilot may advance only after a stratified survey reaches 80/100, no resident group is materially behind, and slow-mobility and event-day tests pass. [source:BEIJING-15-MINUTE-LIFE-CIRCLE] [metric:resident\_transport\_satisfaction\_index]

The park catalogue describes a 24-hour open park with no public external parking area. This proposal therefore does not make a new core parking facility the precondition for events; it prioritises rail, walking, cycling, accessible transfers, peripheral shared parking and reservation management. The complete evaluation protocol, sample, thresholds and failure fallback are in report/narrative.md. [source:JINGZHANG-PARK-CATALOG]

Transport and municipal depth are constrained by [depth:traffic\_rail\_slow\_parking] and [depth:municipal\_new\_infrastructure]. Public-space evidence is [data:geometry/public\_space.geojson#PUBLIC-001].

The constraints layer [data:geometry/constraints.geojson#CONSTRAINTS] records the pending interfaces. Road red lines, station entrances, utilities, fire access, parking and event-day capacity remain pending inputs; the strategy is not written as an approved conclusion.

Public services are human-first. Health navigation is non-diagnostic; enterprise service uses authorised materials and human review; data demonstrations stay in a sandbox; and a network outage switches to printed guidance, staffed counters and ordinary public operations.

The operating matrix places AI industry and innovation services, talent life, new infrastructure, distributed energy, edge computing and conventional municipal systems in one service framework. Each interface records a spatial position, user, accountable operator, service radius and phasing dependency. Utilities, energy, drainage, flood control and fire-safety inputs are incomplete and remain prerequisites for formal deepening; the proposal therefore describes reversible service interfaces, not engineering commitments.

## Urban Resilience, Embodied Intelligence, and Full-Life-Cycle Iteration

The future city is treated as a coupled human–environment–machine–asset system, not a list of smart objects. The climate-adaptation evidence boundary covers heatwaves, cold snaps, cloudbursts, drought, thunderstorms and freeze–thaw; air and microclimate cover ventilation, heat island, shade, PM2.5, dust and allergens; water and ecology cover terrain, pipes, infiltration, groundwater, water quality, biodiversity, bird collision and dark skies; human experience covers older people, children, disabled and pregnant people, night workers, noise, lighting, cognitive load, service access and belonging; embodied intelligence covers mixed traffic, emergency stop, tele-operation, charging/fire separation, information security, appeal and accountability; lifelines cover power, water, drainage, heat, communications and emergency response; construction covers reuse, materials, energy, carbon and waste; and operations cover asset ownership, inspections, desilting, spares, budgets, skills, work orders, renewal and exit. [source:RESILIENT-CITY-INFRASTRUCTURE-2024] [source:BEIJING-ACCESSIBILITY-REGULATION] [source:BEIJING-BIRD-BIODIVERSITY-2024] [assumption:A-BIODIVERSITY-LIGHT-001]

### Low-regret comparison across all states

Five explicit design prototypes were compared: S0 minimal change, S1 technology-intensive, S2 ecology-intensive, S3 event-capacity-intensive and S4 balanced adaptation. The model uses 11 dimensions—thermal comfort, stormwater, air, human experience, robots, privacy, biodiversity, maintenance, carbon, flexibility and cost efficiency—five stakeholder weight profiles and eight stress states. A fixed seed 147228 produced 50,000 Monte Carlo draws with uncertainty in both weights and indicator responses. These are normalized decision scores, not field performance, climate forecasts or approval criteria. [assumption:A-RESILIENCE-MCDA-001] [metric:resilience\_v13\_candidate\_count] [metric:resilience\_v13\_monte\_carlo\_draws]

|                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| · Prototype · Main advantage · Main cost or failure mode · Mean robust score · P05 · Win rate ·                                                                                                                |
| · --- · --- · --- · ---; · ---; · ---; ·                                                                                                                                                                       |
| · S0 minimal change · Low capital intensity and high reuse · Weak stormwater, maintenance, mixed-mobility and heat performance · 42.559 · 40.365 · 0% ·                                                        |
| · S1 technology-intensive · Strong robot testing and digital base · Privacy, ecology, operational complexity and offline degradation · 54.090 · 50.613 · 0% ·                                                  |
| · S2 ecology-intensive · Best stormwater, biodiversity, heat and carbon response · Higher event-capacity, separation and everyday maintenance pressure · 70.242 · 66.812 · 21.138% ·                           |
| · S3 event-capacity-intensive · High event and logistics throughput · Hardscape, noise, heat, drainage and habitat costs · 46.579 · 42.994 · 0% ·                                                              |
| · S4 balanced adaptation · Human experience, robot safety, maintenance, privacy and flexibility pass together · Higher capital intensity; S2 wins under a pure climate weighting · 72.518 · 70.252 · 78.862% · |

S4 has mean regret of 0.314 and the lowest design score across eight stresses is 67.194. It passes five pre-set screening gates: core minimum ≥60, stress minimum ≥55, capital intensity ≤85, privacy ≥80 and human experience ≥75. S2 still wins for the climate-priority profile; that counterexample matters because S4 is a cross-group robust compromise, not the strongest option on every single dimension. [metric:resilience\_v13\_selected\_mean\_score] [metric:resilience\_v13\_selected\_p05\_score] [metric:resilience\_v13\_selected\_win\_rate] [metric:resilience\_v13\_selected\_mean\_regret] [metric:resilience\_v13\_selected\_min\_stress\_score] [metric:resilience\_v13\_hard\_gates\_passed]

The eight stresses are a +2°C heatwave, +20% cloudburst, 30% irrigation reduction, 24-hour network outage, a threefold increase in robots, sudden event crowding, a 20% maintenance-capability reduction and winter freeze–thaw. Formal refinement must replace normalized inputs with seasonal CFD, measured wind/temperature/humidity/mean-radiant temperature, terrain and pipe models, infiltration, water quality, acoustic and light surveys, ecology baseline, energy, asset cost and real user tests. [assumption:A-CLIMATE-STRESS-001] [assumption:A-AIR-WIND-001] [assumption:A-DRAINAGE-SYSTEM-001]

### S4 spatial operating system

- Wind, heat and air: keep seasonal ventilation connections along the belt and three branches; do not form a fixed wind wall with continuous buildings, screens or dense planting. Compare leaf-on/leaf-off, prevailing wind, calm-pollution and heatwave cases before fixing trees, buildings or event structures. The city-scale ventilation network is background principle only, not an official Jing-Zhang corridor. [source:BEIJING-VENTILATION-NETWORK-2035] [source:BEIJING-METEOROLOGICAL-OPEN-DATA]
- Rain and drainage: use the five-part chain source reduction → local detention → pipe connection → exceedance conveyance → post-storm recovery. The Haidian public sponge-city range of 75–80% annual runoff control is a screening reference only; final values require sub-area, terrain, soil, pipe, outfall and basement evidence. Underground space is never treated as a storm refuge. [source:HAIDIAN-SPONGE-CITY-PLAN] [source:BEIJING-FLOOD-PLAN-2021-2025]
- The body first: a continuous step-free spine, heat refuge chain and quiet night chain have priority over exhibitions, delivery and robot charging. Older people, children, wheelchair/cane/guide-dog users, prams, cyclists and night workers must join route tests; a model score cannot overrule lived discomfort or a complaint. [source:BEIJING-ACCESSIBILITY-REGULATION] [source:WHO-HEAT-HEALTH-2026] [assumption:A-HUMAN-EXPERIENCE-001]
- Embodied-intelligence levels: progress from world-model simulation to closed test site, low-speed scheduled geofence and only then limited public test. Identity, emergency stop, human takeover, appeal and withdrawal remain visible. Physical safety and GB/T 45502-2025 information security are both required; residents and disabled people are not free robot-training data. [source:BEIJING-EMBODIED-INTELLIGENCE-2025-2027] [source:SERVICE-ROBOT-INFOSEC-GB45502] [source:ISO-TR-4448-PUBLIC-MOBILE-ROBOTS] [assumption:A-EMBODIED-AI-SAFETY-001]
- Ecology and night: use native, layered and drought-tolerant habitat, linked by ecological stepping stones. Review bird collision at glass interfaces; sensitive segments use dark-sky, low-glare, time-controlled light. No species increase or net ecological gain is claimed before a baseline survey. [source:BEIJING-BIRD-BIODIVERSITY-2024] [source:BEIJING-LIGHTING-GUIDE-2025] [assumption:A-BIODIVERSITY-LIGHT-001]

6. Maintenance is design: every tree pit, inlet, permeable surface, seat, light, sign, sensor and robot bay receives an asset ID, owner, inspection trigger, spare, service window, manual fallback and exit route before construction. Two consecutive overdue work orders or unavailable spares trigger removal of complex equipment in favour of passive, standardised, replaceable parts. [source:ASSET-MANAGEMENT-GBT33172] [source:RESILIENT-CITY-INFRASTRUCTURE-2024] [assumption:A-LIFECYCLE-MAINTENANCE-001]

Wind, water and health: a cultural boundary with an engineering gate

Cang feng ju qi and feng shui may appear as cultural landscape narratives, never as medical causation, air-quality evidence, hydrological law, engineering model or approval criterion. The current local baselines remain unknown and must be split into auditable gates:

- pedestrian-level wind-comfort acceptable-area ratio, tested for seasonal directions, calm conditions and leaf-on/leaf-off with calibrated CFD and field wind measurements; [metric:pedestrian\_wind\_comfort\_acceptable\_area\_ratio] [source:LIU-URBAN-VENTILATION-2017]
- pollutant-stagnation hotspot count, requiring an emissions inventory, PM2.5 observations, boundary conditions and a professionally accepted air-age or ventilation-effectiveness method; [metric:pollutant\_stagnation\_hotspot\_count] [source:MENG-WIND-HEAT-PM25-2022] [source:NOSEK-STREET-CANYON-2025]
- mean radiant temperature and continuous shaded accessible-route ratio, requiring seasonal solar/tree-canopy modelling, MRT measurement and accompanied-route tests; [metric:mean\_radiant\_temperature\_baseline\_c] [metric:continuous\_shaded\_accessible\_route\_ratio]
- blue-green accessible-route overlap, requiring field verification of entrances, gradients, continuity, safety, water quality and maintenance, without inferring health benefit from proximity; [metric:blue\_green\_accessible\_route\_overlap\_ratio] [source:WHO-URBAN-HEALTH-AND-GREEN]
- water-risk exceedance-route verification, requiring DEM, drainage, outfalls, soils, groundwater, water quality, design storms and calibrated surface-flow modelling before any flood-free statement. [metric:water\_risk\_exceedance\_route\_verified\_ratio] [assumption:A-DRAINAGE-SYSTEM-001]

design\_target in visual/assets/evidence-ledger.json is a future review gate; the six unknown entries in metrics.json are the current evidence state. They may become known only when simulation input, calibration, field observation, formula, error and a responsible professional are all documented. wind-health-validation-plan.json and wind-health-field-protocol.json bind every observation to a geometry version, boundary case, source schedule, point ID, instrument, height, wind, PM2.5, thermal variables, canopy state and QC flag. Missing geometry alignment, calibration, safe field conditions, source/detection-limit information or pre-registered sampling stops interpretation and keeps the metric unknown. The identified Haidian 1+21+65+100 network and Qinghe Station wind-monitoring route are lawful acquisition and context routes only, not site observations. [source:AIJ-CFD-PEDESTRIAN-WIND-2008] [source:AIJ-CFD-GUIDEBOOK] [source:ISO-7726-INSTRUMENTS-2025] [source:HAIDIAN-METEOROLOGICAL-NETWORK-2023] [source:QINGHE-STATION-WIND-MONITORING-2021]

Key-area stress tests and trigger-based rollback

- Zhongzhi Garden: treat the Qinghe edge as a shared wind, water, low-carbon-component and closed embodied-intelligence test; check drainage, winter slip, robot failure and data security before display.
- AI Origin Community: prioritise shade, quiet night learning, walk/cycle continuity and everyday service; robots operate only in windows that do not occupy the accessible spine.
- Dazhongsi: resolve the station's four quadrants, cloudburst ponding, event crowds, ride-hailing/delivery, night noise and post-event recovery; event capacity yields to residents returning home, fire safety and exceedance conveyance.

Triggers convert future change into an executable fallback: heat warning reduces event capacity and opens refuge points; heavy-rain warning clears low points, checks inlets and opens exceedance routes; network loss stops autonomous robot movement while public facilities stay in manual mode; one non-takeover event, one blocked accessible spine or one serious near miss pauses robot testing; an asset without an owner or two overdue maintenance cycles blocks expansion. These are design triggers, not forecasts or statutory thresholds.

Stricter Monte Carlo robustness check

To avoid making the proposal look artificially strong under fixed weights, a separate S0–S4 experiment samples five design prototypes under five stakeholder-weight profiles, eight stress states and score noise. It uses 50,000 deterministic Monte Carlo draws with random seed 147228. [metric:resilience\_v13\_candidate\_count] [metric:resilience\_v13\_monte\_carlo\_draws]

The S4 balanced-adaptation option has an approximately 78.862% win rate, mean regret of 0.314 points, a robust mean score of 72.518 and a P05 score of 70.252. These are comparative design results, not field performance. [metric:resilience\_v13\_selected\_mean\_score] [metric:resilience\_v13\_selected\_p05\_score] [metric:resilience\_v13\_selected\_win\_rate]

The minimum item across the eight stresses and the gate state are recorded in [metric:resilience\_v13\_selected\_mean\_regret], [metric:resilience\_v13\_selected\_min\_stress\_score] and [metric:resilience\_v13\_hard\_gates\_passed]. They still require calibration against site data.

The next optimisation direction is therefore explicit: add passive shade and maintainable stormwater storage; reduce dependence on high sensor counts; improve offline operation and human takeover; decouple event capacity from quiet space; and establish asset redundancy and replacement-part standards. The Beijing public meteorological catalogue is a route for later climate calibration, but the Haidian download currently requires a platform user key, so unacquired temperature, rainfall, wind-speed and humidity values are not presented as facts. [source:BEIJING-METEOROLOGICAL-OPEN-DATA] [assumption:A-METEOROLOGICAL-DATA-001]

Drainage targets must still be reconciled with the published Haidian sponge-city planning context. The ventilation network is background direction at city scale; the provisional spine must not be called an official ventilation corridor. [source:HAIDIAN-SPONGE-CITY-PLAN] [source:BEIJING-VENTILATION-NETWORK-2035] [assumption:A-DRAINAGE-SYSTEM-001] [assumption:A-AIR-WIND-001]

The next embodied-intelligence validation covers the Beijing embodied-intelligence action plan, service-robot information-security standard and full-life-cycle smart-municipal-infrastructure requirements. A robot must pass data security, offline operation, human takeover and responsibility-traceability tests in addition to physical obstacle avoidance. [source:BEIJING-EMBODIED-INTELLIGENCE-2025-2027] [source:SERVICE-ROBOT-INFOSEC-GB45502] [source:RESILIENT-CITY-INFRASTRUCTURE-2024] [assumption:A-EMBODIED-AI-SAFETY-001]

Maintenance and ecology are checked against asset-management, bird-friendly and Beijing night-lighting references. Critical facilities need an accountable chain and service-life record; trees, glass and lighting need bird and dark-sky baselines. The proposal does not present any species count, glass-collision rate or night illuminance as an existing-condition conclusion. [source:ASSET-MANAGEMENT-GBT33172] [source:BEIJING-BIRD-BIODIVERSITY-2024] [source:BEIJING-LIGHTING-GUIDE-2025] [assumption:A-LIFECYCLE-MAINTENANCE-001] [assumption:A-BIODIVERSITY-LIGHT-001]

Cross-evidence keeps human health, public participation and sustainable operation in one risk table: WHO and IPCC materials indicate that air, noise, heat, blue-green space, mobility safety and extreme rainfall require coordinated checks. [source:WHO-URBAN-HEALTH-AND-GREEN] [source:IPCC-AR6-URBAN-RISK]

The human-centred intelligence frameworks from NIST and UN-Habitat require explainability, participation, digital rights, equity, interoperability, budgets and continuous monitoring. Resident surveys, accessible-route tests, complaints/appeals and annual review cannot be replaced by robot efficiency. [source:NIST-HUMAN-CENTERED-AI] [source:UN-HABITAT-PEOPLE-CENTRED-SMART-CITIES]

The Beijing walking/cycling standard provides a local interface for continuity, public-space and street-maintenance checks. Water reports provide interfaces for warning, storage, pre-rain clearing, mixed-connection remediation and emergency dispatch; road-maintenance guidance connects inspection, asset ledgers, flood-season response, bridge/tunnel equipment and emergency handling into the operating loop. [source:BEIJING-WALK-CYCLE-DB11-1761] [source:BEIJING-WATER-REPORT-2024] [source:BEIJING-ROAD-MAINTENANCE-2026]

The embodied-intelligence boundary also uses ISO 13482 for mobile/assistive-robot hazard reduction and ISO 55001:2024 for full-life-cycle asset management. The former informs testing, emergency stop, takeover and physical human–machine risk; the latter informs asset performance, risk, expenditure, operation, maintenance and continuous improvement. Neither is a deployment permit or a local procurement standard. [source:ISO-13482-SERVICE-ROBOT-SAFETY] [source:ISO-55001-2024]

All-State Urban Operating System and Director's 30-Second Film

The all-state matrix tests the same public route under clear weather, rain, heat, night, network loss, power loss, sensor drift and failed robot takeover. In every state, tactile and printed guidance, emergency stop, lighting, help and staffed operation retain a minimum service. Embodied intelligence passes four gates—edge-compute, embodied-AI governance, privacy minimisation and a model card—before a small pilot; autonomy rate never replaces public access. Water follows source infiltration → local detention → pipe discharge → exceedance conveyance. Trees, winter slip, dismantlable materials, microgrid degradation, maintenance debt, asset responsibility and annual failure drills sit in one lifecycle loop. The dashboard distinguishes known, design\_target, unknown and blocked; a blocked accessible spine, unsafe rain low point, failed emergency stop, unclear privacy boundary or overdue maintenance automatically degrades to manual mode and stops expansion. [source:NIST-HUMAN-CENTERED-AI] [source:ISO-13482-SERVICE-ROBOT-SAFETY] [source:ISO-55001-2024] [source:WHO-URBAN-HEALTH-AND-GREEN]

To keep “someone is responsible” from remaining a slogan, visual/assets/operational-assurance-contract.json makes each test window register its scenario and geometry version, operating/asset/emergency roles, ordinary equivalent service, cyber isolation and patch contact, liability and insurance assumption, redress path, flood-equity check, independent review and public incident summary. Four assurance tracks cover embodied intelligence, event/public space, blue-green drainage and data/model security; each has evidence, stop conditions and rollback. This is a conceptual review contract, not an insurance policy, legal opinion, construction permit, measured flood model or cybersecurity certification. Any unknown or blocked item is a no-go. [data:visual/assets/operational-assurance-contract.json]

The short film is a communication layer, not engineering proof. Its five shots are deliberately traceable:

- 0–6 s, morning wind: a low tracking shot follows tree shadow, a tactile rail mark and a resident's wheelchair route; sound is leaves, not a synthetic “smart-city” swell. Overlay: known / geometry version / no-app route.
- 6–12 s, rain arrives: macro water runs from permeable paving into a rain-tree pit; a maintenance tag and inlet inspection appear, while a low point remains visibly reserved for exceedance flow. Overlay: design\_target / drainage gate / post-storm work order.
- 12–18 s, station crossing: at Dazhongsi's four-quadrant interface, an older resident crosses first, a delivery worker slows, and the status board shows a booked window. Overlay: human priority / step-free chain / public stop.
- 18–24 s, outage: the network indicator drops; the robot brakes and a human spotter takes over. Printed wayfinding and a staffed counter continue. Overlay: blocked AI layer / ordinary service continues.
- 24–30 s, night review: low-glare light, a quiet seat, an open contribution wall and a maintenance work order close the loop; the final frame says scale, redesign or retire. Overlay: receipt / rights / evidence / rollback.

Each shot maps to visual/assets/evidence-ledger.json#film-storyboard and #film-production, the relevant geometry layer, a sound cue, copyright record, operating state and fallback. The film must not imply official construction, measured health benefit, robot certification or a guaranteed future. [data:visual/assets/evidence-ledger.json#film-storyboard]

## Blue-Green Network, Public Space, and Urban Character

The blue-green loop is a climate adaptation and social infrastructure, not a decorative green ribbon. Shade, quiet seats, permeable edges, rainwater storage, low-glare lighting and accessible routes are designed together. A component can fail digitally without failing physically: the rainwater tree pit still drains, the tactile route still works, and the seat remains a seat.

The identity system is two parallel rails, three open nodes and two switch diamonds. It turns the railway memory into a contemporary public grammar without using un-cleared historical images. The palette and wayfinding rules, including high contrast, tactile and audible alternatives, are machine-readable in visual/assets/identity-system.json.

This iteration also makes the identity direction inspectable as a vector mark in assets/identity/open-pulse-mark.svg: two parallel rails, three open nodes, two switch diamonds and bilingual naming remain legible at different scales. It is a concept asset authored by Xu Bingnan / Codex, not a registered trademark, government emblem or finished VI system; trademark, font, accessibility and public-communications review remain required before use.



English translation of figure text

The source graphic is preserved above; this panel translates its detected text.

Identity system · two rails / three nodes / two switches  
Construction rule: two parallel rails · three public nodes · two switch diamonds.  
Public components: status board — what is booked / who is accountable; stitch threshold — step-free, audible, tactile crossing; rain tree pit — shade + storage + work order; withdrawal bay — robot stops without blocking people.  
Palette: rail graphite, public jade, open blue, civic amber, human coral.  
Concept direction · no registered mark · every active component requires a passive human equivalent and a withdrawal path.

## Renewal Projects, Implementation Policy, and Phasing

This submission defines six renewal projects and registers their locations, types, functions, accountable parties, dependencies, phases, risks, evaluation indicators, phasing ranges and drawing links in operations-matrix.json, construction-readiness.json and compliance\_matrix.json. It separates [metric:phase\_count] conditional phase bands from [metric:constraint\_gate\_count] data-gap gates: the former express when a later window may become possible, while the latter identify interfaces still requiring official boundary, ownership, road/rail, water, fire and accessibility evidence. Policy interfaces cover urban-renewal coordination, spatial provision, operations, industry services, public participation, data governance and property-rights coordination; phasing.geojson expresses a conceptual phase sequence, not an approved construction schedule.

Project-list and phasing depth are managed by [depth:renewal\_project\_list] and [depth:phasing\_implementation], with spatial evidence in [data:geometry/phasing.geojson#PHASE-001]. Without ownership, funding, an implementation body and an approval route, the proposal records an implementation risk rather than a delivery promise.

· Project · Project name · Type · Main dependency · Evidence ·

· --- · --- · --- · --- · --- ·

· JZ-01 · Jing-Zhang Heritage Park slow-mobility stitching · Public space / transport · Road red lines, under-bridge space and traffic-organisation review · [data:geometry/roads.geojson#ROAD-001] ·

· JZ-02 · Zhongzhi Garden Qinghe innovation edge · Blue-green space / industry display · River blue line, ecology and flood-control conditions · [data:geometry/green\_space.geojson#GREEN-001] ·

· JZ-03 · Origin Community near-campus transfer street · Urban renewal / industry service · Campus boundary, ownership and ground-floor uses · [data:geometry/buildings.geojson#BLDG-001] ·

· JZ-04 · Dazhongsi Station four-quadrant pedestrian continuity · Transit integration / slow mobility · Rail station, road intersections and municipal utilities · [data:geometry/public\_space.geojson#PUBLIC-001] ·

· JZ-05 · AI public-service and edge-compute nodes · New infrastructure / public service · Energy, compute, security and operating body · [data:geometry/constraints.geojson#CONSTRAINTS] ·

· JZ-06 · Global AI Week public route · Operations / identity · Public-space permission, event safety and rights clearance · [data:geometry/phasing.geojson#PHASE-001] ·

The 100-day open-call design period is kept separate from urban-renewal implementation phases. The 90-day sequence is only a conditional arrangement for locking data, preparing the base, testing passive components and opening one bounded window; it is not a government timetable. Near-term work can begin with physical wayfinding, paper maps, staffed counters and low-disturbance public services. Longer-term renewal, event routes, developer communities and international communication must wait for official regulatory plans, municipal, transport, ownership, permission and accountability evidence. operations-matrix.json reserves review fields for operating objects, frequency, responsible party, conversion route, risk and exit rules.

Implementation is a sequence of reversible windows rather than a one-off technology launch:

1. Read and prepare: confirm official polygons, survey accessibility, drainage and heritage; publish the rights ledger and baseline.
2. Pilot and learn: open low-risk public-space components and bounded scenario windows with a staffed operator; the four industry-validation windows remain small, reversible and subject to separate release gates.
3. Connect: add the regional return loop, enterprise and research interfaces only after the stop gates pass.
4. Scale or retire: publish evidence, compare equity and climate outcomes, and either expand, redesign or remove the intervention.

The scenario-operation-matrix.json uses relative low, medium and high cost bands only; they are not budgets or procurement estimates. RACI and SLA fields identify the future owner needed for a formal implementation plan. If a responsible operator, safety case, rights record, drainage check or non-AI equivalent is missing, the scenario cannot advance.

## Metrics, Area Recalculation, and Compliance Matrix

The current quantitative decision experiment compares baseline, adaptive, regenerative and stress states. adaptive is the preferred candidate within this internal low-regret comparison; relative model scores are not competition scores, causal benefits, engineering costs or government commitments. The historical and 2030 raw extrapolations are diagnostic only: rate-like values outside 0–100 or intensity-like values below 0 are flagged model outputs, not public targets, existing facts or scoring inputs. The complete raw series, targets, backtest and fallback actions remain in visual/assets/model-backtest.json.

The preferred indicators are district income per capita, district GDP per capita and district health institutions per 10,000 people, with green coverage, physicians, wastewater treatment and water intensity as guardrails. Every number must be read with its source, year, unit, scope and limitation in visual/assets/evidence-ledger.json, visual/assets/model-backtest.json, visual/assets/indicator-selection.json and metrics.json.

Once official polygons replace the provisional geometry, the package must rerun area, ratios, key-area counts, object counts, accessible-route checks, blue-green continuity, drainage interfaces, land-use coverage, constraint interfaces and phasing. A passing self-check is evidence of package integrity, not evidence that a provisional concept is legally approved.

## Risk, Copyright, and Compliance

The package includes a path-level visual/assets/copyright-ledger.json and an expanded report/copyright\_statement.md. Text, geometry, diagrams, offline HTML and figures are authored or derived in this worktree by Xu Bingnan / Codex. Official public data retain their source attribution. No third-party images, maps or remote runtime assets are embedded; local system fonts are used only for rendering and are not redistributed. The identity mark is a concept direction, not a registered trademark.

The rights ledger records authorship, source inputs, license scope, hashes and review limits for each asset. It is not a legal opinion. Any future team member adding an image, font, map, logo, data extract or event content must add a source, permission and hash before publication. Historical interpretation is expressed with text and self-generated diagrams unless a rights-cleared source is explicitly registered.

Public-interest checks cover accessibility, older residents, carers, night workers, privacy, conflict resolution and ordinary non-AI access. The package rejects facial recognition and individual trajectory retention for these scenarios. The stop condition is part of the design, not an afterthought: a failed safety, rights, drainage, access or resident-impact gate triggers human operation or withdrawal.

### Spatial commitment: verifiable public value before an AI spectacle

The proposal commits to an auditable public route, ordinary service and reversible release rather than an unbounded technology display.

### Three key areas: design moves for professional deepening

Each key area remains a provisional concept with a specific spatial move and a professional gate for geometry, ownership, fire, heritage, ecology, drainage and municipal review.

### Fourteen scenario cards and governance boundaries

The fourteen cards bind a trigger, spatial carrier, minimum data, human responsibility, ordinary equivalent and stop condition; they are not service-level or deployment commitments.

### Four industry validation windows: evidence before procurement or scale-up

The validation windows ask for reproducible safety, rights, human-review and rollback evidence before procurement or expansion.

### Cultural narrative, public markers, and long-term operations

Heritage interpretation, public markers and seasonal operations stay conceptual until rights, public-interest, maintenance and event-clearance gates pass.

## v1.6 Official Statistical Time Series and Decision Experiment

The v1.6 time-series section is retained in the Chinese source document and its machine-readable assets. It records the official statistical source boundary, the 2023 baseline frame and the four scenario comparison. It does not turn modeled outcomes into official forecasts. Readers should use the time-series figure and ledger to inspect source years, transformations, missing values and guardrails.

# v1.7 Evidence-to-Delivery Layer: From Concept to Reviewable Deliverables

This section presents the package's evidence-to-delivery layer:

- visual/assets/taskbook-crosswalk.json gives every agent.1–agent.6 requirement one unique section, artifact and acceptance test.
- visual/assets/regional-ecosystem.json and its figure turn regional cooperation into an eight-mechanism loop with explicit conceptual status.
- visual/assets/scenario-operation-matrix.json turns 14 scenes into spatial, operational and rollback records with RACI, SLA, cost bands and non-AI equivalents.
- visual/assets/component-library.json gives eight physical, legible and maintainable public-space components.
- visual/assets/identity-system.json and the identity figure provide a real, inspectable identity and wayfinding grammar.
- visual/assets/copyright-ledger.json makes authorship, sources, licenses, hashes and future-clearance duties auditable.
- visual/assets/case-mechanism-matrix.json, key-area-node-plans.json and public-interest-audit.json close the case-comparison, node-level and participation/equity evidence gaps without inventing a baseline or partnership.
- visual/assets/operations-matrix.json turns the design into eight accountable action packages—from official-boundary replacement and field baselines to embodied-intelligence windows, drainage maintenance and annual scale/redesign/retire decisions.
- visual/assets/resource-accounts.json makes five dependencies auditable: spatial rights, climate/water/ecology, model/data rights, human experience/participation, and assets/spares/maintenance. These are resource requirements, not budgets or funding commitments.
- visual/assets/construction-readiness.json adds five construction-to-opening gates and four passive/connected component interfaces; it does not add a construction drawing, quantity, budget or approval claim.
- visual/assets/participation-cycle.json records the return-pass process, participant lenses, current known/unknown states and the evidence required before a public window; it does not represent resident consent or a completed consultation.
- proposal.md, this English translation, two offline report pages, the offline visual index and changelog.md keep the bilingual review surface aligned.

These are reviewable design deliverables, not claims that a project has been commissioned. Their value is that a reviewer can find the requirement, inspect the evidence, test the stop gate and identify what must be confirmed before implementation.

The local review-surface record is visual/assets/qa-readiness.json. It records semantic HTML tables, locally linked bilingual figures and A3/A0 outputs, the offline-only runtime boundary, and the known limits that provisional geometry and unknown field baselines still require professional and public review.

The implementation handoff is deliberately inspectable: operations-matrix.json gives each package an owner, accountable reviewer, dependency, acceptance test, stop condition, maintenance rule and relative cost band; resource-accounts.json states which evidence and capacity must exist before a package is released. Together with the scenario matrix and the Open Pulse test-window record, these files form a reviewable chain from a public question to a bounded window, a human handoff, an evidence record and a scale/redesign/retire decision.

## 1. Taskbook index and regional cooperation

visual/assets/taskbook-crosswalk.json binds each agent.1–agent.6 requirement to a unique section, artifact and acceptance test. Regional links remain conceptual and returnable rather than partnership claims.

## 2. Fourteen scenario–space–operations matrix

Each scenario row carries a spatial carrier, trigger, minimum data, RACI, recovery rule, non-AI equivalent and exit condition.

This revision adds visual/assets/run-open-pulse-scenario-audit.js and its persisted result open-pulse-scenario-audit.json. The runner reads only the bundled 14 scenarios and eight operation packages, then deterministically checks the spatial carrier, trigger, minimum data, RACI, ordinary fallback, stop/success conditions, acceptance, maintenance and provisional boundary. Three deliberately malformed fixtures—missing accountable role, missing ordinary fallback/stop rule, and missing operation stop rule—must be rejected. node visual/assets/run-open-pulse-scenario-audit.js --check proves only that the structural contract is reproducible; it does not prove field access, operator SLA, model safety, engineering approval or a public pilot.

### 3.1 Node-level concepts and public-interest audit

The node plans and public-interest audit make arrival, status, operation, ordinary return, accessibility, complaint and group-difference evidence reviewable.

## 3. Identity system, public-space components, and embodied intelligence

The identity grammar and component library require offline human service, visible emergency stop, maintenance ownership and safe degradation before any embodied-intelligence window.

## 4. Copyright, privacy, and public-interest clearance

Path-level hashes, rights boundaries, data minimisation, withdrawal and ordinary access remain release conditions.

### 4.1 Rights and build provenance: itemised evidence

The clearance and build record puts auditable facts in the proposal, sources.json and the path-level ledger: authorship, inputs, licence boundary and SHA-256 return to [source:BUILD-PROVENANCE-OPEN-PULSE]. This is package self-declaration, not a third-party licence or legal opinion.

- Drawing output: drawings/a3-booklet.pdf reads as 10 A3 pages and drawings/a0-boards.pdf as 5 portrait A0 boards; both carry package author metadata. This does not claim print certification, font licensing or government publication approval. [source:PDF-OUTPUT-PROFILE-OPEN-PULSE]
- Font boundary: CJK text is rendered with local system fonts; no font file is redistributed and no local renderer font is presented as a granted licence. Font embedding, reader display and an authorised publication font remain a pre-publication check. [source:FONT-RENDERING-BOUNDARY-OPEN-PULSE]
- Generation and checks: text, GeoJSON, metrics, figures, offline HTML and PDFs are authored or derived in this submission worktree. Manifest and ledger hashes can be recomputed from package files; self\_check.json records package integrity only and does not claim that build scripts, field results or official scoring are shipped. [source:VALIDATION-TRACE-OPEN-PULSE]

## 5. Bilingual review and version verification

The Chinese master, this translation, local HTML, figures and drawings share one version and evidence ledger; a translation never adds an official fact.

## Construction readiness and continuous participation

The package now separates “ready to discuss” from “ready to build or open”. visual/assets/construction-readiness.json applies five gates: evidence/site lock; a public-route and passive-component mock-up; wind, heat, water and lifeline coordination; a bounded, staffed and time-limited window; and a public retain/redesign/retire decision [metric:construction\_readiness\_gate\_count]. Each gate names entry evidence, outputs, pass and stop conditions, an ordinary non-AI baseline and a maintenance response. It explicitly rejects construction drawings, statutory plans, procurement specifications, cost estimates and implementation approval.

visual/assets/participation-cycle.json makes the return pass inspectable: synchronize inputs, reread changed requirements, compare peers, validate the package, publish a bounded review draft, establish field and public baselines, run a staffed window, return negative and distributional evidence, then retain, redesign or retire. Eight public-user lenses are intentionally grouped into seven non-identifying participation cohorts; the two counts refer separately to the persona matrix and participation cycle, and both have paper, staffed,



multilingual or no-app routes [metric:participation\_lens\_count]. The current package records repository sync, local validation and peer comparison as known review activities, while field baseline, resident consultation and public pilot remain unknown or not\_started; no consent, microclimate result or health outcome is inferred.

## v1.8 Reviewer-Visible Evidence Layer

The v1.8 revision places the highest-impact taskbook evidence directly in proposal.md so a reviewer can audit it without opening every JSON attachment. The following crosswalk is an exact reading of the taskbook, not a new statutory claim.

· Taskbook layer · Jing-Zhang answer · Reviewable proof ·

· --- · --- · --- ·

· Three positions · Century Jing-Zhang culture belt; urban AI life-experience belt; AI-integrated innovation belt · Two-rail identity, no-app service, research → public test → evidence loop ·

· Five functions · Full-stack autonomy; world-class ecosystem; AI+ scene paradigm; intelligent active city; global AI-governance voice · Zhongzhi Garden test gate, Origin open room, 14 scenarios, Dazhongsi public sequence, red-team and responsibility desk ·

· Three areas / two wings · AI Origin Community; Zhongzhi Garden; Dazhongsi; Zhongguancun technology-service wing; Xiaoyuehe scene-empowerment wing · Provisional key-area geometry, node plans, regional ecology, blue-green and operations records ·

The regional chain explicitly includes the Beijing North Latitude Community / Haidian universities and institutes, Huairou Science City, Future Science City, Beijing E-Town and the Beijing-Tianjin-Hebei network. Each interface has an input, an output, a future owner and evidence required before activation. All are conceptual suggestions; no partnership, funding, land allocation or government commitment is asserted.

· Regional interface · Input · Jing-Zhang output · Responsibility or evidence to confirm ·

· --- · --- · --- · --- ·

· North Latitude Community / Haidian universities · Resident questions, talent, open-source topics · Public scenario briefs, participation feedback, reproducible experiments · Community consent, participation support, privacy boundary ·

· Huairou Science City · Upstream science and facility capacity · Explainable public displays and validation questions · Facility operator consent and data clearance ·

· Future Science City · Materials, energy and engineering tests · Low-carbon components, drainage and robot-interface prototypes · Engineering, ecology and municipal review ·

· Beijing E-Town · Manufacturing, industrialisation and deployment · Public-validation evidence after safety gates · Enterprise/park cooperation, procurement and liability boundary ·

· Beijing-Tianjin-Hebei network · Diverse urban contexts and comparative feedback · Diffusion, reuse or retirement records · Regional authorities and operators ·

### 1. Hard mapping of three positions, five functions, and three areas/two wings

The taskbook positions, functions and spatial structure are mapped to the identity system, no-app service, three nodes and the regional return loop.

### 2. Regional cooperation as a returnable responsibility chain

Each regional interface has an input, Jing-Zhang return, future owner and evidence required before activation; none is a partnership or land-allocation promise.

### 3. Key-area node-level plan

Each node follows arrival → public status → component → bounded window → ordinary public use, with a professional gate before refinement.

### Node and operations table

· Node · Human sequence · Embodied-intelligence boundary · Gate and acceptance test ·

· --- · --- · --- · --- ·

· Zhongzhi Garden · arrival → status board → stitch threshold → rain tree pit → booked test → park return · low-speed, booked window; visible emergency stop; never blocks the accessible chain · river/flood/traffic/ownership/fire review; a resident crosses, reads and leaves without an account ·

· AI Origin Community · campus edge → open room → rights desk → contribution wall → learning → quiet residential edge · contributions can be attributed or withdrawn; data stays in a sandbox · parcel/heritage/displacement review; an older resident can use the route and service offline ·

· Dazhongsi · rail arrival → four-quadrant crossing → quiet seat → short exhibition → international route → daily retail · event capacity can fall to zero; public broadcast and withdrawal bay remain visible · station/road/fire/crowd review; the route remains open and the space returns to ordinary use ·

The 14-row matrix gives every scenario a spatial carrier, trigger, minimum data, RACI, response/recovery SLA, relative cost band, retention, non-AI equivalent and stop condition. In review shorthand: the walking and accessibility group pauses on a broken route; robot testing exits on a serious conflict or failed stop; red-team testing rejects unclear provenance; public publishing removes uncleared material; health navigation never diagnoses; events reduce to zero if fire, quiet or resident-access gates fail; rain and quiet-chain pilots freeze on overdue drains or threshold breaches.

· Scenario group · Spatial carrier · Trigger and minimum data · Responsibility and recovery gate · Non-AI equivalent / exit ·

· --- · --- · --- · --- · --- ·

· Walking-gap diagnosis and accessibility assistance · Roads and public space · Manual counts, obstacle log, accessible route · Transport/accessibility lead; seasonal check and 7–14 day complaint response · Human guide and physical signs; stop if the primary chain breaks ·

· Low-speed delivery robot · Road constraints and withdrawal bay · Geofence, emergency stop, spotter, event log · Operator responsible, safety lead accountable; 24-hour review · Human delivery/handcart; exit after severe conflict or failed stop ·

· Model red-team and data sandbox · Controlled test room and status board · Cleared sample, model card, evaluation log · Evaluation team and lab operator; publish window summary · Paper review/expert panel; reject unclear provenance ·

· Open publishing, enterprise service and data room · Origin Community/service node · Authorised materials, public process, audit log · Community/service lead; 7-day record and correction route · Staffed counter and notice board; fuse on withdrawal ·

· Health navigation and cultural wayfinding · Community node and green space · Non-diagnostic flow, cleared history, tactile/high-contrast guide · Service/park lead; staffed window always available · Human explanation, paper and audio; remove on diagnostic risk or unclear source ·

· Intelligent retail and global AI week · Dazhongsi commercial sequence · Aggregate crowd, fire, quiet zone and event content · Event control/culture lead; 30-minute return and 14-day review · Ordinary retail/park opening; reduce capacity or cancel on threshold breach ·

· Rain/tree-pit and quiet night chain · Green space, drainage and public space · Pre/post-rain ponding, light/noise log, manual inspection · Asset lead; 24-hour pre-rain and 48-hour post-rain closure · Manual clearing and physical wayfinding; freeze on overdue inlet or threshold breach ·

### Public-interest and case-comparison evidence

The baseline is deliberately unknown until surveyed. Older residents, disabled people, carers, night workers, children and guardians, visitors, small businesses and maintenance crews are named participants. Step-free continuity, service without an app, complaint response, group differences and night light/noise are audited through paired walks, staffed tests, stratified listening and public status boards. A conflict pauses the scenario, shows the disputed evidence, offers a non-AI equivalent and publishes remedy or withdrawal.

